You (in)complete me:

Partial devoicing and vowel duration in a large corpus of Dutch

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Introduction

Problématique

- Our theories are only as good as the data that inform them are representative.
- As tools get more sophisticated and data get more massive and more available, there is a natural temptation to applying **big data** to linguistic problems, as a complement to laboratory experiments or field studies.
- Of course, these sources pose their own unique problems, and there is the perennial question of when “big” is “too big.”

Today's talk

- In this presentation, we look at **incomplete neutralization** of final devoicing in Dutch in a small subset of articles from the Spoken Wikipedia project.
- This study is part of a larger project aiming eventually to offer tools to linguists looking to exploit this data source in any of the languages supported.
- Specifically, today we will be focusing on the general methodology and what it yields on a (*very!*) preliminary dataset.
- We tentatively find further evidence that this neutralization is indeed incomplete in Dutch – however, our effects mostly go in the **opposite direction** of those of the literature.
- Despite its obstacles, an undertaking like this seems feasible and worthwhile!

Roadmap

Background

Dutch inventory

- Dutch has the following obstruent phonemes (Booij, 1995):

(1)	Bilabial		Labiodental		Alveolar		Velar		Glottal
	p	b			t	d	k		
			f	v	s	z	x	ɣ	ɦ

- [g] exists only in loanwords, and palatalization before /j/ turns /s, z, t/ to [ʃ, ʒ, c].
- The voicing distinction in pairs /f/ – /v/ and /x/ – /ɣ/ is being lost in Standard Dutch in favour of the voiceless member word-initially and intervocalically.

Voicing & devoicing in Dutch

- Dutch famously displays **final devoicing** (e.g., Van Oostendorp, 2008). Voiced obstruents are not permitted in syllable-final position, where they surface as voiceless.

(2)	/bɛd/	/bɛdən/	/bɛt/	/bɛtən/
	[bɛt]	[bɛdən]	[bɛt]	[bɛtən]
	'bed'	'beds'	'(I) wet'	'(we) wet'

- Dutch also has two processes affecting voicing in clusters (Booij, 1995):
  - **Progressive devoicing** targets fricatives after voiceless obstruents (e.g., *o*/pv/*allend* 'remarkable' → [pf]). Final devoicing on the first member can feed this.
  - **Regressive voicing** targets voiceless obstruents before voiced stops (e.g., *o*/pd/*ruk* 'imprint' → [bd]). This leads to superficial exceptions to final devoicing.
- Some exceptions occur (Grijzenhout and Krämer, 2000):
  - Obstruents resyllabify and retain their underlying voicing before most affixes. However, devoicing accompanies resyllabification at the end of a prosodic word (e.g., /roːd+ɑxtəx/ → [((roːt)$_{\omega}$ ɑxtəx)$_{PPh}$] 'reddish').
  - Regressive voicing does not apply across certain morpheme boundaries, such as the past tense; instead, the past tense allomorph [-tə] is selected over typical [-də] (e.g., [stɔp] 'stop', [stɔp-tə] 'stopped').

Incomplete neutralization

- Neutralization processes like devoicing in Dutch are presumed to eliminate their underlying distinction and induce a categorical change (e.g., [t] $\leftarrow$ /d/ should be functionally indistinguishable from [t] $\leftarrow$ /t/).
- However, research has found evidence for **incomplete neutralization**, where the outputs of neutralization retain some indices of their underlying features:
  - Devoiced obstruents show significant differences from underlyingly voiceless ones with respect to closure voicing, burst duration, and/or preceding vowel length in German (e.g., Port and O'Dell, 1985), Polish (e.g., Slowiaczek and Dinnsen, 1985), Catalan (Dinnsen and Charles-Luce, 1984) and Russian (e.g., Dmitrieva, 2005).
  - Similar effects have been found for other neutralized properties, such as flapping, aspiration and tone (cf. Braver, 2019 for references).
- Typically, these effects are diminished in the (incompletely) neutralizing environment but go in the same sense. For instance, vowels are longer before underlyingly voiced obstruents than voiceless, but the difference is much greater in contrastive environments (Warner et al., 2004).
- Speakers in some studies are even able to perceive the difference between these two types of obstruents (e.g., Port and O'Dell, 1985).
- Not all neutralization processes are supposed to be incomplete, though. For example, Kopkallı (1993) finds no significant effect in Turkish devoicing.

What about Dutch?

- Warner et al. (2004) find evidence for incomplete neutralization in Dutch final devoicing:
  - Vowels are significantly longer before /d/ than /t/ in neutralizing environments, by approx. 3.5 ms, regardless of their phonemic length.
  - Burst duration was significantly different after long vowels depending on underlying voicing (/t/ > /d/ by approx. 9 ms).
  - Underlying voicing was not significant with respect to closure duration (/d/ vs. /t/), and closure voicing duration was only significant in contrastive environments.
- Follow-up work, however, finds that these effects are modulated by orthography alone, and they conclude there is no incomplete neutralization in Dutch (Warner et al., 2006).
- Similar results have been found for German (Fourakis and Iverson, 1984) and Polish (Jassem and Richter, 1989).
- Meanwhile, though Ernestus and Baayen (2006) find effects of orthography on final devoicing in Dutch, other production and perceptual results in their experiments lead them to conclude that “incomplete neutralization is part and parcel of the grammar of Dutch” (p. 46).

Spoken Wikipedia

- The **Spoken Wikipedia** project is a crowd-sourced collection of recordings of Wikipedia articles. Any editor can contribute.
- 27 languages are represented, although some are more abundant than others (e.g., Portuguese has only 90 articles).
- Previous work on this corpus (for German, English and Dutch) has been done by Köhn et al. (2016). Our work distinguishes itself with its goals to:
  - Provide a methodology and scripts for working with any language of the corpus;
  - Perform alignment using the Montreal Forced Aligner (McAuliffe et al., 2017) over MAUS (Kisler et al., 2017), as the former has been shown to be more accurate (Gonzalez et al., 2020);
  - Offer both phonetic- and phonological-level transcription; and
  - Generate files natively compatible with Praat (instead of XML files).
- In addition to the recordings and the article text at the time of recording, information on speakers is sometimes available through the recording's WikiMedia commons page and/or the speaker's user page, though the latter does not have a standardized format.

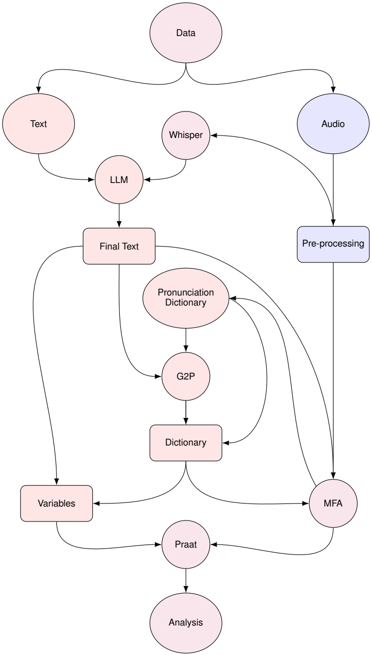
Methodology

Principles

- The idea is to be as modular and multi-language as possible from the beginning; language-specific resources don't interest us as much.
- The approach presented here is a general overview for the Spoken Wikipedia project and should apply to any language represented there.
- Languages best represented by this workflow are those for which the MFA has an existing dictionary and acoustic model, but these could be built or trained from other sources with a few extra steps.
- Generalizing a bit, this approach could be adapted to any data source comprising of matched audio & text.

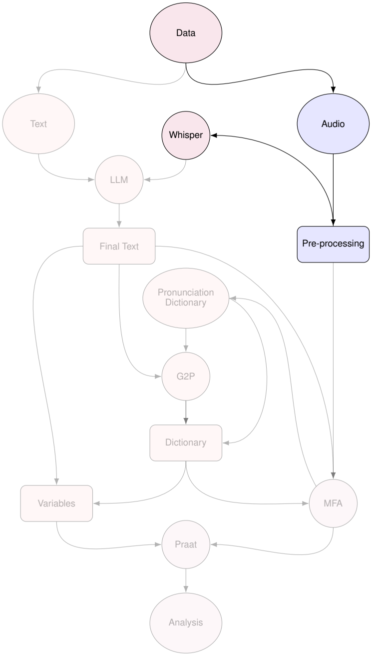
Complete workflow

- We scrape the language-specific Spoken Wikipedia page for the audio links, article text (version at the time of recording, if specified) and attempt to gather speaker information (not included here).
- Legend:
  - Data sources = ellipses; Programs/tools = circles; Intermediate products = boxes
  - Text sources = red; audio sources = blue; both = purple



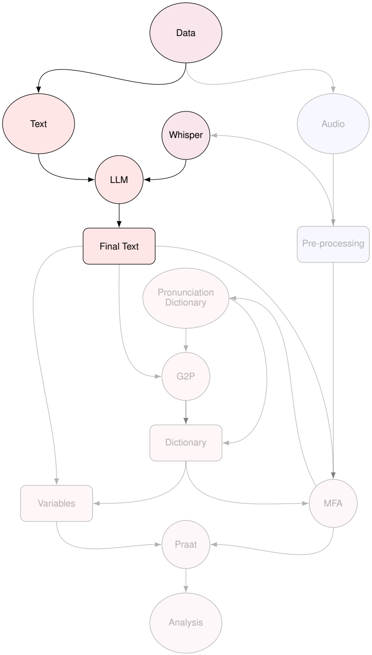
Workflow: Audio processing

- We **pre-process** the audio files, namely .ogg to .wav conversion and bitrate normalization.
- Transcription of audio is then performed with **Whisper** using the `faster-whisper` package.



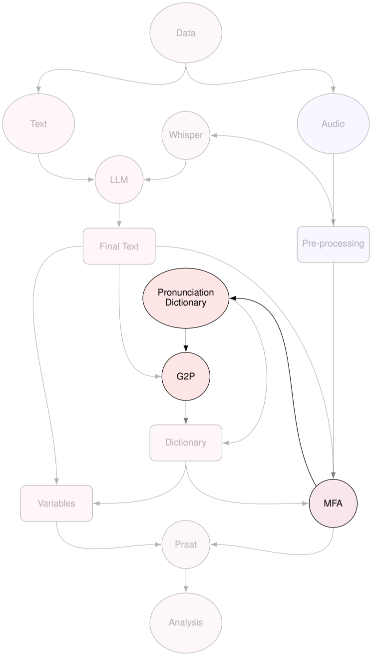
Workflow: Text processing

- An **LLM** (gpt-4o) reconciles the original article text with the **Whisper** transcription, prioritizing the latter (see Appendix 1 for prompt).
- Digits (cardinal and ordinal) are converted to spelling using `num2words` package.
- This gives us a **Final Text**.



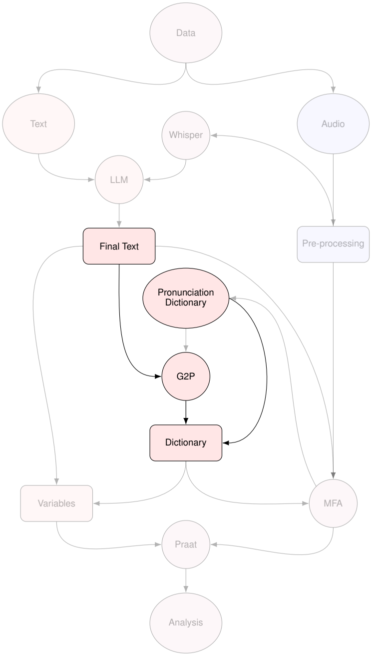
Workflow: G2P model

- When available, the **MFA** provides a **Pronunciation Dictionary**.
- We train a Grapheme-to-Phoneme model (**G2P**) model within the **MFA** based on the **Pronunciation Dictionary**.



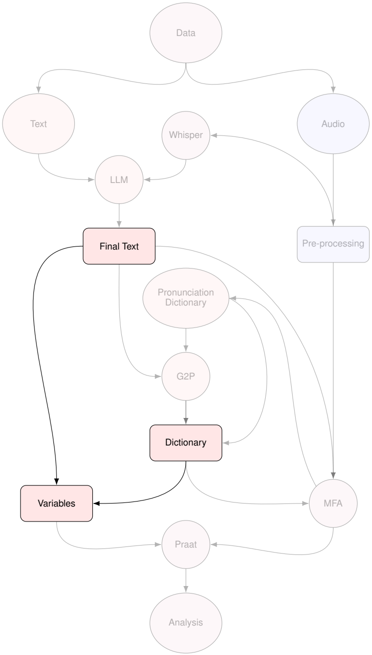
Workflow: Corpus dictionary

- The words in the **Pronunciation Dictionary**, plus the missing transcriptions estimated by the **G2P**, give us a working **Dictionary**.



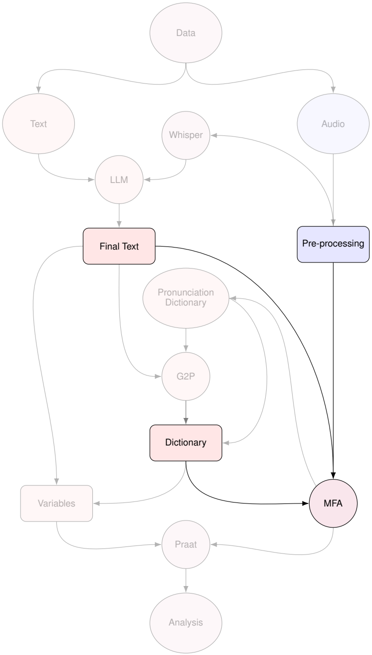
Workflow: Variables

- Using the transcriptions in the **Dictionary** as well as the position of words in the **Final Text** (i.e., for following or preceding words), we identify **Variables** of interest.
- Note that these **Pronunciation Dictionaries** tend to favour superficial variants (e.g., words like /bɛd/ are given as [bɛt] and *maybe* [bɛd] as well).



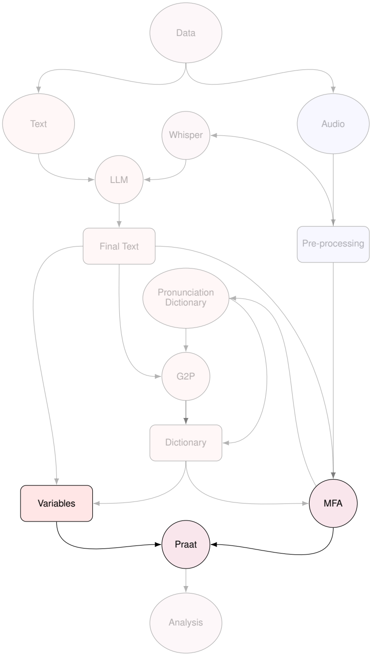
Workflow: Forced alignment

- The **Final Texts** and their accompanying **pre-processed audio** are provided to the **MFA** using its pre-trained acoustic model.
- The modified **Dictionary** is also provided to the **MFA**.



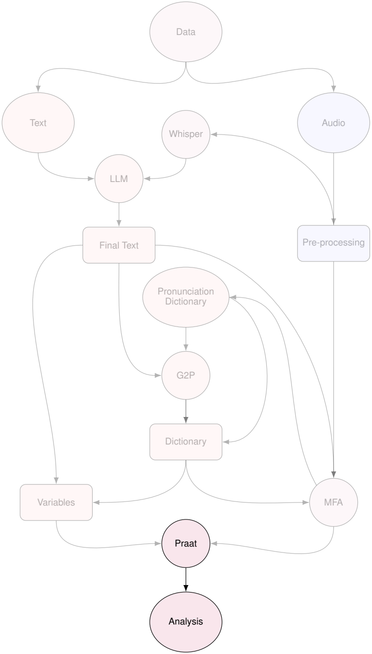
Workflow: Importing to Praat

- Audio and verified TextGrids from the MFA are provided to **Praat**.
- A version of the TextGrids targeting our **Variables** of interest is generated, when necessary.



Workflow: Getting measurements

- A **Praat** script takes the relevant measurements.
- And we're ready to process our data and do some **Analysis**!



Going Dutch

What's specific to today's presentation?

- Low-frequency noise was present in some of the files, so we had to make a separate, de-noised version of the audio for voicing analysis with `ffmpeg` using a 100 Hz high-pass filter with a 12 dB/octave rolloff.
- We took a random sample of 97 of 100 articles (article link missing for 3 recordings), for approx. 5.5 hours of speech.
- We used a fine-tuned Dutch Whisper model: **hannatoenbreker/whisper-dutch**.
- Voicing information was extracted (= number of unvoiced frames for each interval from the Voice Report function) from our high-pass filtered sound files using a modified Praat script from Al-Tamimi and Khattab (2015).
- We extracted word and segment labels, interval numbers, duration and voicing percentage for each interval on the segment tier.
- Since the pronunciation dictionary and the G2P model were based on *phonetic* transcriptions, we had to jerry-rig a matching scheme to deduce from the orthography whether a surface voiceless obstruent came from its voiced counterpart or not.

Tokens

- For each target, we extracted the label of the preceding and following labels, along with the voicing percentage of these intervals where non-empty.
- Target (underlying) sounds and environments (where L = sonorant, C = obstruent, V = vowel):
  - **Sounds:** /p, b, t, d, k/
  - **Onset environments:**
    - #_{C, L, V} (e.g., **p**sychische, **b**redevoort, **p**ijfelijk)
    - {C, L, V}_V (e.g., aans**p**oelden, olym**p**ische, o**p**eratie)
    - {C, L, V}_L (e.g., terug**b**rengen, buiten**p**laatsen, pro**b**leem)
  - **Coda environments:**
    - {L, V}_# (e.g., keukenhul**p**, verkoo**p**)
    - {L, V}_C (e.g., sym**p**tomen, o**p**geroepen)
- **Total tokens:** 30,313

A word of warning

- Only a cursory verification of a few TextGrids has been done, to make sure the aligner wasn't going completely off the rails.
- Our sample has only 10 speakers, one of whom contributed **74%** of the recordings. This seems pretty indicative of the corpus as a whole.
  - This speaker is 41 years old,
  - Presumably male,
  - Born in Putten, Netherlands (about 65km from Amsterdam) and
  - Works as an equipment manufacturer.
- We are not performing any tests of statistical significance at this point.
- Most of the results presented today focus on word-final and word-initial position.

Results

Sanity check: [+son, +cons] voicing by context

Context	Mean
#_V	84.8
C_V	75.7
V_V	96.6
L_V	99.1

Table: Onsets

Context	Mean
V_#	88.3
V_C	93.8
V_L	98.8
L_#	86.0
L_C	95.0

Table: Codas

Note that the voicing of sonorants in the C_V context is much higher when we separate based on the superficial voicing of the previous obstruent.

Obstruent voicing percentage by underlying [voice]

Onset, Voiced

Context	Mean
#_C	68.6
#_V	58.4
#_L	47.9
C_V	45.6
C_L	38.3
V_V	81.7
V_L	72.0
L_V	82.5
L_L	67.8

Onset, Voiceless

Context	Mean
#_C	30.7
#_V	14.8
#_L	16.3
C_V	8.80
C_L	8.43
V_V	38.0
V_L	47.6
L_V	26.1
L_L	33.5

Coda, Voiced

Context	Mean
V_#	37.8
V_C	79.6
L_#	31.6
L_C	68.2

Coda, Voiceless

Context	Mean
V_#	53.9
V_C	62.1
L_#	34.7
L_C	57.2

What about word boundaries?

- The MFA provides an empty interval if it detects a pause. We used this to determine whether word-final stops were followed by a pause or another word, accounting for whether its initial segment was a consonant or a vowel.
- Even then, underlyingly voiced stops showed less voicing on average than voiceless stops, regardless of the following environment.

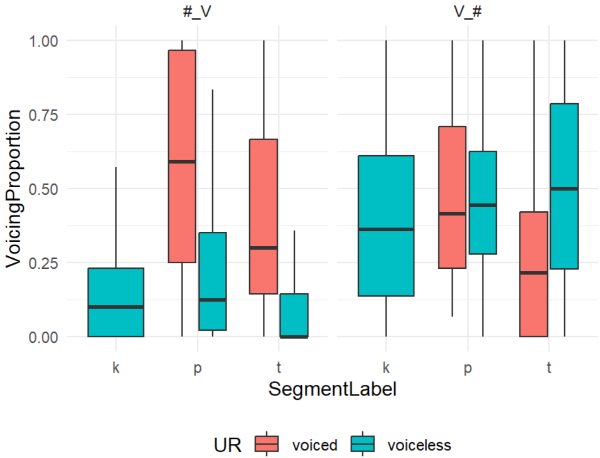


Figure: Voicing by underlying [voice], place and context

- Voicing percentage in word-final position is roughly equal between /b/ and /p/, but smaller for /d/ than /t/.
- There is obviously considerable variation and some suspicious findings that warrant further inspection.

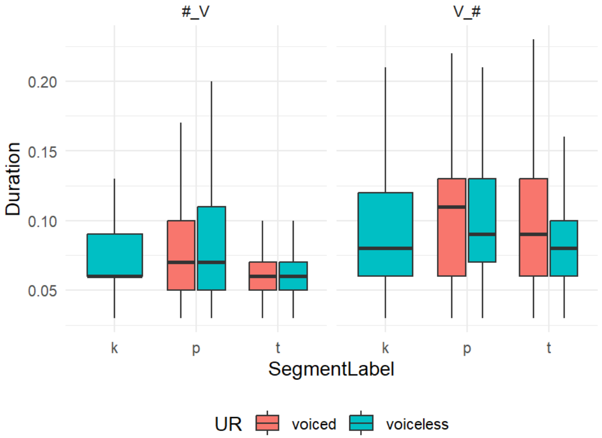


Figure: Stop duration by underlying [voice], place and context

- Stops are longer in word-final position, but no clear distinction *vis-à-vis* place or underlying voicing.

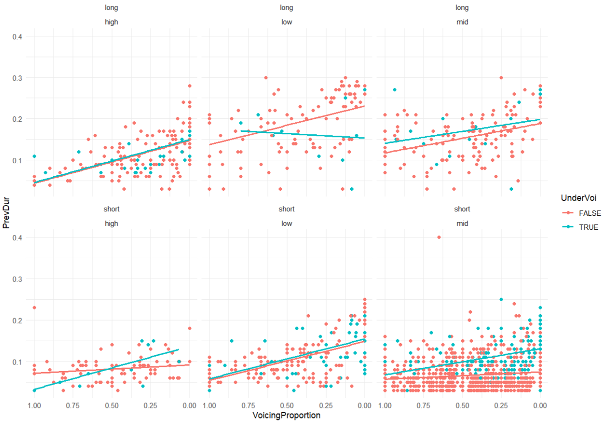


Figure: Preceding vowel duration vs. stop voicing in word-final position, by underlying voicing, vowel height & length

- Generally, the less voiced a stop is in word-final position, the longer its preceding vowel is.
- This effect is *sometimes* more pronounced when the stop is underlyingly voiced.
- An informal look at the interaction of place with voicing suggests place does not play a role, though the inverse trade-off of voicing and length is quite pronounced for /aːt#/ sequences.

Discussion & conclusion

Discussion

- Our results suggest some corroborating evidence for **sub-phonemic distinctions** between underlyingly voiced and voiceless stops in Dutch.
- Stop duration and voicing for word-final labials (/b/ = /p/) seem not to discriminate by underlying [voice].
- Coda /d/ distinguishes itself from coda /t/ in that it is on average *less* voiced, while in between onset /d/ and onset /t/.
- Finally, in the word-final position, there is an effect of vowel length on preceding vowels which may interact with underlying obstruent voicing.
- However, the trade-off goes in the opposite sense. We would expect phonetic vowel length to increase proportionately with voicing (cf. Kluender et al., 1988 and references therein).

Conclusion

- Our results remain, of course, preliminary and are contingent on manual verification.
- Some theory of deliberately increasing the salience of devoiced obstruents may offer some motivation, but should these trends hold, they could pose a problem for formal accounts of incomplete neutralization.
- For instance, a consequence of Braver's (2019) Weighted Paradigm Uniformity theory of constraints is that neutralized candidates showing values greater than those of their counterparts are harmonically bounded.
- In the table below, Braver compares a lengthened short vowel (reference in contrastive environments: 50 ms) to an underlyingly long vowel (reference: 150 ms).
- Crucially, there is *no* pair of weight values which can lead to an optimal total cost when the length of the candidate surpasses that of the long counterpart's reference duration.

Lengthened V dur (ms)	*cost*(OO-ID(dur)) $w_{\mathrm{ID}}(Dur(cand) - Dur(base))^2$	*cost*(DUR($\mu\mu$)=TARGDUR($\mu\mu$)) $w_{\mu\mu}(TargDur(\mu\mu) - Dur(\mu\mu))^2$	*total cost*
75	$1 \times (75-50)^2$	$3 \times (150-75)^2$	17500
100	$1 \times (100-50)^2$	$3 \times (150-100)^2$	10000
125	$1 \times (125-50)^2$	$3 \times (150-125)^2$	7500
150	$1 \times (150-50)^2$	$3 \times (150-150)^2$	10000

Figure: Costs for lengthening, where $w_{\mathrm{id}} = 1$, $w_{\mu\mu} = 3$, $TargetDur(\mu) = 50$ ms & $TargetDur(\mu\mu) = 150$ ms.

Moving forward

- We are looking into extracting more information in keeping with the literature, such as burst and closure duration of stops, as well as more sophisticated and/or multi-faceted measurements of voicing.
- Morphology needs to be looked at more closely; currently, word types that are known to be exceptional in Dutch (such as compounds) are mixed in.
- Working with neutralization is very tricky with these sorts of tools!
    - In the eventual absence of *phonemic* pronunciation dictionaries, we may need to generate our own using a set of modified computational rules *à la* **XPF Corpus** (i.e., no devoicing).
    - These transcriptions would *replace* the MFA's pronouncing dictionary when training our G2P.
    - Application (complete or not) or inapplication could then be deduced from the phonetic values themselves and/or from context.

Thanks

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Appendix 1: Prompt

You are a text transformation assistant. Your goal is to generate an accurate transcript of an audio recording that read part of a Wikipedia article. You are given two texts:

1. **The original Wikipedia article text** (which may contain references or sections not read)
2. **An automatic transcript from Whisper** (which may contain mistakes, especially with proper nouns)

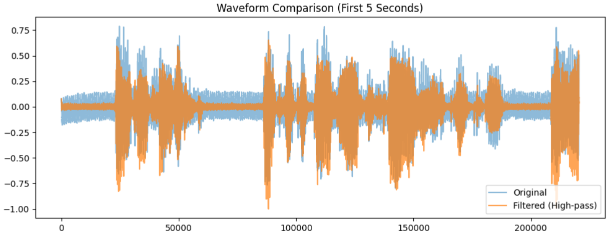
Your task is to generate a **cleaned transcript** that exactly matches what was spoken, following these guidelines:

- **Prioritize the Whisper transcript** as the primary source. If there is uncertainty about a phrase, prefer what appears in Whisper.
- **Correct errors** in the Whisper text (e.g., proper names, numbers, and grammar), using the Wikipedia text as a reference, but **do not introduce missing information** from Wikipedia unless it is clearly necessary to fix a mistake.
- **Preserve section headers** only if they appear in the Whisper transcript.
- **Include the preamble and epilogue** from Whisper, as they are part of the audio.
- **Maintain the order and structure** of the Whisper text as much as possible.
- **Do not add additional section headers** that were not in Whisper.
- **Convert cardinal digits** to their spoken equivalents.
- **Convert ordinal digits** to their spoken equivalents. (For instance, since we are dealing with Dutch, `10e` should be turned into `tiende`.)

Example:

Whisper	Tot een elletje kwam het in 1999, toen gezing door het IOC berispt werd wegen het aannemen van vijfduizend dollar van de olimpische organisatie in Salt Lake City, kandidaat voor de winterspelen van twee duizend twee.
Wikipedia	Tot een relletje kwam het in 1999, toen Geesink door het IOC berispt werd wegens het aannemen van 5000 dollar van de Olympische organisatie van Salt Lake City, kandidaat voor de Winterspelen van 2002.
Expected Output	Tot een relletje kwam het in duizendnegenhonderdnegenennegentig, toen Geesink door het IOC berispt werd wegens het aannemen van vijfduizend dollar van de Olympische organisatie in Salt Lake City, kandidaat voor de Winterspelen van tweeduizendtwee.

Final Rule: When in doubt, **keep the Whisper phrasing**, making only essential corrections. Do not introduce words or phrases from Wikipedia unless necessary for accuracy.


Waveform Comparison (First 5 Seconds)
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
50000
100000
150000
200000
Original
Filtered (High-pass)